

Stock Sentiments Analysis: Legacy News Media vs. Reddit

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Abstract

This study investigates the relationship between sentiment extracted from legacy news media and Reddit post titles and its impact on stock price movements for six high-volume stocks (AAPL, META, NVDA, PLTR, TSLA, SPY) from March 2024 to April 2025. Utilizing large language models for sentiment classification, we construct sentiment indexes and analyze their correlation with stock prices through Cross-Correlation, Lead-Lag Correlation, and Granger Causality Tests. Our findings reveal that news sentiment, particularly data-driven and fiscal-related categories, exhibits stronger and more consistent correlations with stock price movements compared to Reddit sentiment, which often lags behind price changes. Although the Granger Causality Tests reveal limited predictive ability from both sentiment indexes, machine learning models that incorporate both types of sentiment indexes demonstrate enhanced performance in predicting stock price directions. Additionally, our backtested trading strategy based on our models achieved a 1.123% return during a challenging period, outperforming baseline strategies. These results highlight the value of sentiment indexes as complementary features for stock prediction models.

1 Introduction

The Fear and Greed Index is a popular sentiment index that reflects the stock market's sentiment. It helps investors measure whether the market is fairly valued or emotions drive the stock prices. The range of emotions is from extreme fear to extreme greed. This index is calculated based on the movement of the stock market, option market, future market and bond market. When investors are fearful, the stock market may be undervalued, signalling potential buying opportunities. On the other hand, the stock market may be overpriced if investors are very greedy with their investment decisions. Therefore, sentiment analysis of stocks has become an essential tool for investors and analysts.

Media, including social media, can significantly influence and drive public sentiment. Traditionally, legacy news media, such as Bloomberg, CNBC, and the Wall Street Journal, have been the primary sources of public financial information. Institutional reporters and analysts regularly post financial news and analyses on those platforms. However, in recent years, retail investors have begun sharing their thoughts and investment decisions directly on social media platforms like Reddit, particularly in communities like r/WallStreetBets and r/stocks. Therefore, retail investors' investment decisions may be influenced by other peers on social media platforms. On the other hand, institutional investors have been instructed to follow a set of studies, analyses, procedures and rules for investment decisions.

In the past, institutional investors accounted for 80% of the trading volume of the New York Stock Exchange (NYSE), while retail investors made up a smaller percentage. Traditionally, buying or selling stocks requires personal brokers to facilitate the trades, which is inconvenient and may involve high management fees. Nowadays, the advancements in financial technology (fintech) have lowered the bar for retail investors to participate in the market. Retail investors can now trade stocks and derivatives directly in real-time using a smartphone device. By 2021, retail investors comprised 25% of the total trading volume, nearly double the percentage reported a decade ago, and the trend continues. Existing works [1, 4, 9] primarily focus on either news or social media sentiment analysis. However, given the evolution of both legacy media and social media, we aim to analyze how sentiment trends differ between these two sources and how these differences impact stock price movements.

In this study, we develop stock sentiment indexes using large language models (LLMs) to classify and quantify sentiment from both news headlines and Reddit post titles. Our findings reveal several key points:

- (1) The sentiment expressed in news headlines and Reddit post titles is highly correlated.
- (2) The sentiment indexes tend to lag behind stock price movements by one day.
- (3) Data-driven and fiscal-related news headlines and Reddit post titles show a stronger correlation with stock price movements compared to opinion-based.
- (4) Both the news and Reddit sentiment indexes serve as valuable features for stock prediction models.

This work aims to provide insights into the evolving landscape of financial information and its impact on investor decision-making.

2 Related Work

Sentiment analysis in the stock market has greatly improved due to advancements in natural language processing (NLP). Transformer-based models like BERTs [3] excel in sentiment analysis by capturing bidirectional context in financial texts. Recently, open-source models like Deepseek [2] have made advanced large language models (LLMs) more accessible. As a result, we aim to explore the potential of Deepseek models for conducting sentiment analysis.

Data sources are crucial for conducting sentiment analysis. Traditional sources, such as financial news and regulatory filings, have been studied extensively. For instance, Fazlija and Harder [4] analyzed articles from Reuters and Bloomberg to identify sentiment trends that often preceded market shifts. Additionally, social media platforms, particularly Twitter (Bollen et al. [1]) and Reddit (Zhang et al. [9]), offer real-time sentiment analysis, indicating that they can effectively predict stock returns and market volatility. However, to the best of our knowledge, no studies have yet compared

traditional sources like news media with social media platforms like Reddit.

Applications of stock sentiment analysis include stock price prediction, portfolio optimization, and risk management. Models integrating sentiment with technical indicators, as explored by Picasso et al. [7], have improved forecasting accuracy. In this work, we primarily focus on developing our stock sentiment indexes. Our experiments also demonstrate that sentiment alone can yield promising results in stock predictions.

3 Problem Definition

In financial markets, news media and social media like Reddit plays a pivotal role in shaping investor sentiment, which in turn can influence stock prices and market behavior. Therefore, our goal is to create stock sentiment indexes based on news headlines and Reddit post titles.

Given a corpus of news headlines or Reddit post titles:

$$\mathcal{H} = \{h_1, h_2, \dots, h_N\}$$

where h_i is a news headline or Reddit post title.

Then, we have a classifier $f : \mathcal{H} \rightarrow \mathcal{Y}$

$$f(h_i) = y_i \in \{-1, 0, +1\}$$

Here, the label space $\mathcal{Y}$ corresponds to:

- -1: Negative sentiment
- 0: Neutral sentiment
- +1: Positive sentiment

Finally, for a given day t , we compute the daily sentiment score by averaging:

$$s_t = \text{mean}(\mathcal{Y}_t) \quad (1)$$

where $\mathcal{Y}_t = \{y_i | t_i = t\}$ and the resulting sentiment index is $\mathcal{S} = \{s_1, s_2, \dots, s_T\}$.

4 Methodology

In this section, we describe the process of creating our sentiment indexes by using large language models (LLMs) to classify and quantify sentiments from news headlines and Reddit post titles. First, we will compile our own dataset from various online sources. Next, we will outline the procedures for building our news sentiment indexes and Reddit sentiment indexes.

4.1 Data Collection Methodology

To build our dataset $\mathcal{H}$ for stock sentiment analysis, we developed a data collection method for both legacy news media and Reddit in order to gather news headlines and Reddit post titles, respectively.

For legacy news media, many websites have advanced anti-bot detectors to prevent web scraping. Finding a way to bypass these detectors can be costly and time-consuming. Therefore, we decided to focus on websites where we can successfully avoid detection. As a result, the websites we selected for this purpose are <https://www.nasdaq.com> and <https://www.benzinga.com>.

On the other hand, social media platforms also have robust anti-bot detectors, so we can only obtain their data through their APIs.

Each social media platform has different policies and restrictions, and accessing data can sometimes be expensive. Among these platforms, Reddit has relatively loose restrictions. We can scrape the most recent 1,000 posts from each subreddit for free using its API, which enables us to build a sizable dataset for our analysis.

However, we still encounter major challenges regarding legacy and social media. Since all the platforms only display the most recent data on their websites, we are limited to collecting what is currently available. Consequently, our dataset only concentrates on the recent timeline. Therefore, we have restricted our analysis from March 2024 to April 2025. For the scope of the study, we focus on five high trading volume stocks, AAPL, META, NVDA, PLTR, TSLA, and the S&P 500 index, for our analysis. **As a result, our dataset of news headlines and our dataset of Reddit post titles contain 28947 and 37808 records, respectively.**

Lastly, to analyze the stock price movement, we also collect the historical daily stock data $\mathcal{S}_{stock} = \{s_1^{stock}, \dots, s_T^{stock}\}$ from Yahoo! Finance using its API, yfinance, where s_t^{stock} is the daily closing price of the stock.

4.2 Sentiment Scoring Methodology

Once we have constructed our dataset of news headlines $\mathcal{H}_{news}$ and Reddit post titles $\mathcal{H}_{reddit}$, the next step is to obtain sentiment labels for each record for sentiment analysis. However, annotating sentiment labels can be both costly and time-consuming. To address this challenge, we decided to leverage natural language processing (NLP) techniques, specifically large language models (LLM) as a classifier f , for sentiment classification. These models evaluate each news headline and Reddit post title to determine sentiment based on their potential impact on corresponding stock price movements. If a headline or title indicates that the stock price is likely to increase, it is classified as positive. Conversely, if it indicates a likely decrease in stock price, it is classified as negative. If the model assesses that the headline or title has minimal impact on stock price, the sentiment is classified as neutral. Additionally, if the sentiment is uncertain, it is labelled as unknown. For this classification task, we used the Deepseek model [2], specifically `deepseek-r1-distill-qwen-7b`.

The prompt used for classification was:

Prompt: Classify the sentiment of this news headline based on its potential impact on its stock price:

1. positive (likely to increase)
2. negative (likely to decrease)
3. neutral (minimal impact)
4. unknown (uncertain impact)

Headline: {headline}

Answer: {answer}

After obtaining the sentiment labels for the news headlines and Reddit post titles, we need to convert these labels into sentiment scores. We assign a score of +1 for positive, -1 for negative, and 0 for neutral, while filtering out any titles labelled as unknown. We

Table 1: Cross Correlation among News Sentiment, Reddit Sentiment and Stock Price

Stocks	News-Reddit Corr. ($\rho_{n,r}$)	News-Stock Corr. ($\rho_{n,s}$)	Reddit-Stock Corr. ($\rho_{r,s}$)
AAPL	0.28**	0.29**	0.10
META	-0.02	0.07	0.14*
NVDA	0.51**	0.21**	0.27**
PLTR	0.13*	0.10	0.35**
TSLA	0.50**	0.48**	0.36**
SPY	0.33**	0.41**	0.20**

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value $\geq$ 0.05).

then calculate the daily sentiment score by averaging all headlines or titles from each day in Equation 1.

Finally, since the sentiment indexes S_{news} and S_{reddit} can be very volatile, making trend analysis difficult, we apply a moving average to smooth the curve, similar to how we analyze stock price movements S_{stock} .

5 Hypotheses

5.1 Research Hypothesis 1: Correlation Analysis

Different platforms can have different perspectives and sentiments regarding the same financial events. Generally, retail investors may act recklessly, and their emotions are often easily influenced by financial events. In contrast, institutional investors tend to possess strong financial knowledge and respond more prudently. However, institutional investors also face performance pressure that could strongly affect their sentiments. By analyzing the sentiment trends in legacy news media and on social media like Reddit, we can gain better insights into the market reactions of both institutional and retail investors. This section investigates whether sentiment extracted from legacy news media and Reddit can predict short-term stock price movements.

Methodology: Sentiment indexes of six prominent stocks are computed separately for each source (news and Reddit) using natural language processing techniques, as outlined in the previous section. To assess the relationship between the two sentiment time series for each stock, we employ two statistical methods:

5.1.1 Cross Correlation Test. This method measures the strength and direction of the linear relationship between the news sentiment index S_{news} and the Reddit sentiment index S_{reddit} . For each stock, we compute the Pearson correlation coefficient ρ , which ranges from -1 to 1, along with the associated p-value to assess statistical significance.

- H_0 : There is no linear relationship between the news sentiment index S_{news} and the Reddit sentiment index S_{reddit} .
 $\rho_{n,r} = \text{Corr}(S_{news}, S_{reddit}) = 0$
- H_1 : There is a significant linear relationship between the news sentiment index S_{news} and the Reddit sentiment index S_{reddit} .
 $\rho_{n,r} \neq 0$

Table 2: Lead-Lag Correlation between News Sentiment and Reddit Sentiment

Stocks	Direction	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5
AAPL	news→reddit	0.03	0.04*	-0.05**	-0.01	0.03
	reddit→news	0.12**	-0.07**	-0.03*	-0.02	-0.00
META	news→reddit	-0.02	-0.12**	-0.10**	-0.07**	-0.05**
	reddit→news	0.15**	0.06**	0.05**	0.00	0.04*
NVDA	news→reddit	-0.04**	-0.10**	0.04*	0.02	-0.05**
	reddit→news	0.24**	0.02	-0.11**	-0.07**	0.10**
PLTR	news→reddit	0.05**	-0.01	-0.06**	0.13**	-0.01
	reddit→news	0.02	0.09**	0.02	-0.03	-0.06**
TSLA	news→reddit	-0.04**	-0.03	-0.04*	0.00	-0.09**
	reddit→news	0.23**	0.02	-0.03	0.02	-0.07**
SPY	news→reddit	0.07**	0.03	-0.02	0.00	0.07**
	reddit→news	0.12**	0.01	-0.04*	-0.08**	0.04*

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value $\geq$ 0.05).

Key Findings: As shown in the News-Reddit correlation column of Table 1, the cross-correlation analysis reveals that five out of six stocks exhibit statistically significant sentiment alignment between legacy news media and Reddit. NVDA and TSLA show the strongest positive correlations ($\rho_{n,r} = 0.51$ and $\rho_{n,r} = 0.50$, respectively), indicating highly synchronized investor sentiment across platforms. AAPL and SPY also demonstrate substantial positive correlations, while PLTR shows a weaker yet still significant relationship. In contrast, META is the only stock that shows no meaningful correlation ($\rho_{n,r} = -0.02$), suggesting divergent sentiment dynamics between the two sources.

5.1.2 Lead-Lag Correlation Test. This method measures the strength of correlation between two time series at various time shifts within sliding windows, helping identify whether changes in one series tend to precede changes in the other. By examining correlations at different lags, we can infer which signal is potentially leading or lagging in time.

Let $\rho_{n \rightarrow r}^{(t)}(k)$ be the correlation that news sentiment index leads Reddit sentiment index at shift k in window t :

$$\rho_{n \rightarrow r}^{(t)}(k) = \text{Corr}(s_t^{news}, \dots, s_{t+15}^{news}, s_{t+k}^{reddit}, \dots, s_{t+k+15}^{reddit})$$

and let $\rho_{r \rightarrow n}^{(t)}(k)$ be the correlation that Reddit sentiment index leads news sentiment index at shift k in window t :

$$\rho_{r \rightarrow n}^{(t)}(k) = \text{Corr}(s_t^{reddit}, \dots, s_{t+15}^{reddit}, s_{t+k}^{news}, \dots, s_{t+k+15}^{news})$$

for $k \in \{1, 2, 3, 4, 5\}$.

- H_0 : There is no consistent lead-lag relationship at shift k .
 $\text{mean}_t(\rho_{n \rightarrow r}^{(t)}(k)) = \text{mean}_t(\rho_{r \rightarrow n}^{(t)}(k)) = 0$
- H_1 : There is a consistent lead-lag relationship at shift k , indicated by a statistically significant average correlation.
 $\text{mean}_t(\rho_{n \rightarrow r}^{(t)}(k)) \neq 0$ or $\text{mean}_t(\rho_{r \rightarrow n}^{(t)}(k)) \neq 0$

Key Findings: In Table 2, the Lead-Lag Correlation Test reveals that Reddit sentiment consistently leads news sentiment across several stocks, with the strongest and most significant correlations observed at shift 1, particularly for NVDA ($mean_t(\rho_{r \rightarrow n}^{(1)}(k)) = 0.24$) and TSLA ($mean_t(\rho_{r \rightarrow n}^{(1)}(k)) = 0.23$). In contrast, the reverse direction (news leading Reddit) shows weaker or negative correlations, suggesting that Reddit may act as an early signal for sentiment shifts that later appear in news coverage.

Conclusion of Hypothesis 1: The analysis reveals a strong and significant correlation between sentiment on Reddit and traditional news outlets. Notably, it demonstrates that Reddit often reflects shifts in investor sentiment before traditional news does. This lead-lag correlation analysis indicates that Reddit could serve as an early indicator of market sentiment, offering potential value for predicting sentiment trends.

5.2 Research Hypothesis 2: Predictive Power Analysis

Market reactions should guide investment decisions. The sentiment from social media like Reddit tends to be noisy but reveals their real-time feelings about the market. On the other hand, the legacy news media should have a stronger signal to reflect the current market sentiment. Therefore, we can analyze the correlation between sentiment trends and stock movement trends. This section investigates whether sentiment extracted from legacy news media and Reddit can predict short-term stock price movements, and critically, compares the predictive strength of these two sources.

Methodology: To assess the predictive power of the sentiment indexes, we use both visualizations and statistical tests. This combined approach offers complementary perspectives, allowing us to capture both intuitive trends and quantitative significance in the data.

5.2.1 Visualizations. Smoothed sentiment indexes and stock price percentage changes are plotted over the analysis period, using a 5-day moving average, as shown in Figure 1. These plots allow for a visualization of the co-movement between sentiment and price trends, as well as any potential leading or lagging behaviours.

Key Findings: As a representative example, Figure 1 for TSLA compares news sentiment (blue), Reddit sentiment (orange), and TSLA stock price (green) from March 2024 to April 2025, showing mixed fluctuations around the zero mark. **News sentiment and stock price often move together, with peaks and dips aligning** (e.g., around November 2024 and December 2024). On the other hand, **Reddit sentiment shows a weaker correlation with stock price movements**, which is further confirmed by the results presented in Table 1 ($\rho_{n,s} = 0.41$ vs $\rho_{r,s} = 0.20$).

In addition, sentiment and stock price trends were produced for APPL, META, NVDA, PLTR, and SPY in the Appendix Figure 9, 14 and 17, each revealing how the sentiment trends moving align with the stock price. These visual results support the correlation findings and suggest that both sentiment sources contain predictive signals, albeit with varying consistency and volatility.

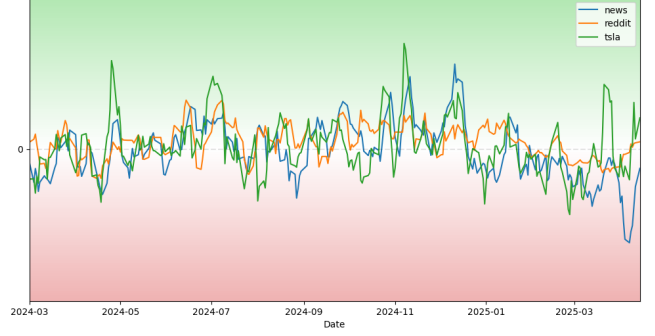


Figure 1: TSLA Sentiment and Price Trends

5.2.2 Lead-Lag Correlation Test. This method measures the strength of correlation between two time series at various time shifts within sliding windows.

- H_0 : There is no consistent lead-lag relationship at shift k .
 $mean_t(\rho_{n \rightarrow s}^{(t)}(k)) = mean_t(\rho_{s \rightarrow n}^{(t)}(k))$
 $= mean_t(\rho_{r \rightarrow s}^{(t)}(k)) = mean_t(\rho_{s \rightarrow r}^{(t)}(k)) = 0$
- H_1 : There is a consistent lead-lag relationship at shift k , indicated by a statistically significant average correlation.
 $mean_t(\rho_{n \rightarrow s}^{(t)}(k)) \neq 0$ or $mean_t(\rho_{s \rightarrow n}^{(t)}(k)) \neq 0$
or $mean_t(\rho_{r \rightarrow s}^{(t)}(k)) \neq 0$ or $mean_t(\rho_{s \rightarrow r}^{(t)}(k)) \neq 0$

Key Findings: Table 3 shows that both the news and Reddit sentiment indexes tend to lag behind stock price movements. Moreover, the correlation is notably stronger in the stock $\rightarrow$ news direction, indicating that news sentiment typically responds to prior changes in stock prices rather than predicting them.

5.2.3 Granger Causality Test. To further investigate the predictive power of the sentiment indexes, Granger Causality Tests determine if one time series can predict another by checking if past values of sentiment indexes (S_{news} or S_{reddit}) improve the prediction of the current value of another series (S_{stock}), beyond using only S_{stock} 's past values. If S_{news} or S_{reddit} 's past values add significant predictive power, S_{news} or S_{reddit} is said to "Granger-cause" S_{stock} .

Let Restricted Model f_k with lag k (uses only past k values of S_{stock}) be:

$$s_t^{stock} = a_0 + a_1 s_{t-1}^{stock} + a_2 s_{t-2}^{stock} + \dots + a_k s_{t-k}^{stock} + \epsilon_t$$

and let Unrestricted Model g_k^{news}/g_k^{reddit} with lag k (uses past k values of S_{stock} and S_{news}/S_{reddit}) be:

$$s_t^{stock} = b_0 + b_1 s_{t-1}^{stock} + \dots + b_k s_{t-k}^{stock} + c_1 s_{t-1}^{news} + \dots + c_k s_{t-k}^{news} + \epsilon_t$$

$$s_t^{stock} = d_0 + d_1 s_{t-1}^{stock} + \dots + d_k s_{t-k}^{stock} + e_1 s_{t-1}^{reddit} + \dots + e_k s_{t-k}^{reddit} + \epsilon_t$$

- H_0 : The unrestricted model g_k^{news}/g_k^{reddit} cannot improve prediction compared to the restricted model f_k . Therefore, the past values of S_{news} or S_{reddit} do not help predict the current value of time series S_{stock} .
(S_{news} or S_{reddit} does not Granger-cause S_{stock})

Table 3: Lead-Lag Correlation between Sentiment Indexes and Stock Price Movements

Stocks	Direction	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5
AAPL	news→stock	0.07*	0.02	-0.06*	-0.11**	-0.07*
	stock→news	0.26**	0.24**	0.20**	0.18**	0.10**
	reddit→stock	-0.06*	-0.11**	-0.17**	-0.21**	-0.27**
	stock→reddit	0.04	0.05	0.07**	0.09**	0.11**
META	news→stock	-0.04	-0.02	-0.06*	-0.01	0.07*
	stock→news	0.13**	0.20**	0.15**	0.17**	0.10**
	reddit→stock	0.13**	0.10**	0.05	0.01	0.01
	stock→reddit	0.17**	0.07**	-0.02	-0.11**	-0.13**
NVDA	news→stock	-0.10**	-0.17**	-0.26**	-0.24**	-0.11**
	stock→news	0.31**	0.36**	0.29**	0.22**	0.16**
	reddit→stock	0.10**	0.01	-0.02	-0.03	-0.06*
	stock→reddit	0.31**	0.30**	0.19**	0.09**	0.00
PLTR	news→stock	0.09**	-0.05*	-0.12**	-0.15**	-0.08*
	stock→news	0.28**	0.28**	0.17**	0.05	-0.02
	reddit→stock	0.17**	0.10**	0.06*	0.04	0.01
	stock→reddit	0.27**	0.25**	0.14**	0.06*	0.02
TSLA	news→stock	0.14**	0.02	-0.13**	-0.20**	-0.17**
	stock→news	0.58**	0.60**	0.49**	0.36**	0.15**
	reddit→stock	0.09**	-0.07**	-0.14**	-0.20**	-0.21**
	stock→reddit	0.33**	0.32**	0.23**	0.11**	-0.05
SPY	news→stock	0.21**	0.05*	-0.13**	-0.29**	-0.28**
	stock→news	0.61**	0.56**	0.42**	0.33**	0.23**
	reddit→stock	0.07**	0.00	-0.09**	-0.16**	-0.18**
	stock→reddit	0.27**	0.26**	0.20**	0.16**	0.08**

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).

- H_1 : The unrestricted model g_k^{news}/g_k^{reddit} significantly improves prediction compared to the restricted model f_k . (S_{news} or S_{reddit} Granger-causes S_{stock})

Key Findings: In Table 4, the Granger Causality Test shows that news sentiment (g_k^{news}) has significant predictive power for stock price movements (f_k) in several cases: AAPL at lag 5, NVDA at lag 1, PLTR at lag 5, TSLA at lag 5, and SPY at lag 2, with p-values below 0.05 or 0.01. In contrast, Reddit sentiment (g_k^{reddit}) shows no significant Granger causality for any stock across all lags, **suggesting news sentiment is a better predictor of stock price movements than Reddit sentiment.**

Conclusion of Hypothesis 2: The analysis reveals that stock price movements consistently lead both news and Reddit sentiment across all examined stocks, as evidenced by the Lead-Lag Correlation Test. On the other hand, the Granger Causality Test indicates that news sentiment has significant predictive power for stock price movements in several cases, while Reddit sentiment shows no predictive ability, highlighting that news sentiment is a more reliable indicator for anticipating stock price changes than Reddit sentiment.

5.3 Research Hypothesis 3: Sentiment Category Impact on Price

This hypothesis explores whether the type of sentiment in financial headlines—*Fiscal-related*, *Data-driven*, or *Opinion-based*—has a differentiated impact on subsequent stock price movements. To analyze the impact of different sentiment types on price changes, we first categorized the sentiments of the headlines. For this classification task, we used the Deepseek model [2], specifically deepseek-r1-distill-qwen-7b.

The prompt used for classification was:

Prompt: You are an AI model that classifies the sentiment type of financial news headlines. Your task is to categorize each headline into one of the following sentiment types:

1. Fiscal-related – Headlines discussing macroeconomic factors, including Federal Reserve decisions, inflation, GDP reports, interest rates, or fiscal policy changes.
2. Data-driven – Headlines based on numerical data, earnings reports, market trends, technical analysis, or statistical insights.
3. Opinion-based – Headlines reflecting personal opinions, analyst predictions, market sentiment, or speculative analysis.
4. Other – Headlines that do not fit into the above categories.

Respond with only the category name: Fiscal-related, Data-driven, Opinion-based, or Other. Do not provide any additional text.

Headline: {headline}

Answer: {answer}

This approach allowed us to group headlines into three primary sentiment categories: *Fiscal-related*, *Data-driven*, and *Opinion-based*. The “Other” category was excluded from further analysis due to lack of relevance to sentiment interpretation.

5.3.1 Cross Correlation Test. This method measures the strength and direction of the linear relationship between the sentiment indexes for each sentiment type S_{news} , S_{reddit} and the stock price movements S_{stock} .

- H_0 : There is no linear relationship between the sentiment indexes for each sentiment type S_{news} , S_{reddit} and the stock price movements S_{stock} .
 $\rho_{n,s} = \rho_{r,s} = 0$
- H_1 : There is a significant linear relationship between the news sentiment indexes for each sentiment type S_{news} , S_{reddit} and the stock price movements S_{stock} .
 $\rho_{n,s} \neq 0$ or $\rho_{r,s} \neq 0$

Key Findings: Table 5 shows varied linear relationships between sentiment types and stock price movements, **with data-driven and fiscal-related sentiments showing significant positive correlations for most stocks, while opinion-based sentiments**

Table 4: P-value of the Granger Causality Test

Stocks	Test	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
AAPL	f_k vs g_k^{news}	0.06	0.17	0.71	0.66	0.00**
	f_k vs g_k^{reddit}	0.21	0.17	0.57	0.74	0.87
META	f_k vs g_k^{news}	0.82	0.85	0.57	0.76	0.71
	f_k vs g_k^{reddit}	0.24	0.32	0.48	0.55	0.67
NVDA	f_k vs g_k^{news}	0.02*	0.06	0.14	0.68	0.09
	f_k vs g_k^{reddit}	0.48	0.66	0.81	0.79	0.29
PLTR	f_k vs g_k^{news}	0.59	0.04	0.09	0.21	0.00**
	f_k vs g_k^{reddit}	0.10	0.17	0.28	0.46	0.41
TSLA	f_k vs g_k^{news}	0.22	0.87	0.06	0.17	0.03*
	f_k vs g_k^{reddit}	0.33	0.49	0.77	0.92	0.93
SPY	f_k vs g_k^{news}	0.27	0.00**	0.37	0.63	0.15
	f_k vs g_k^{reddit}	0.19	0.27	0.77	0.36	0.73

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).

Table 5: Cross Correlation among News Sentiment, Reddit Sentiment and Stock Price for each Sentiment Type

Stocks	Sentiment Type	News-Stock Corr. ($\rho_{n,s}$)	Reddit-Stock Corr. ($\rho_{r,s}$)
AAPL	Data-driven	0.46**	0.08
	Fiscal-related	0.21**	-0.02
	Opinion-based	0.13	-0.31
META	Data-driven	-0.00	0.16*
	Fiscal-related	0.09	0.23**
	Opinion-based	0.09	-0.23
NVDA	Data-driven	0.22**	0.17**
	Fiscal-related	0.12	0.05
	Opinion-based	0.07	0.05
PLTR	Data-driven	0.19**	0.05
	Fiscal-related	-0.04	0.16*
	Opinion-based	0.06	0.18
TSLA	Data-driven	0.43**	0.36**
	Fiscal-related	0.49**	0.38**
	Opinion-based	0.36*	0.19
SPY	Data-driven	0.32**	0.16**
	Fiscal-related	0.31**	0.16**
	Opinion-based	0.08	0.02

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).

often lack significance, indicating stronger market responsiveness to factual sentiment drivers.

5.3.2 Lead-Lag Correlation Test. This method measures the strength of correlation between two time series at various time shifts within sliding windows for each sentiment type.

- H_0 : There is no consistent lead-lag relationship at shift k .
 $mean_t(\rho_{n \rightarrow s}^{(t)}(k)) = mean_t(\rho_{s \rightarrow n}^{(t)}(k))$
 $= mean_t(\rho_{r \rightarrow s}^{(t)}(k)) = mean_t(\rho_{s \rightarrow r}^{(t)}(k)) = 0$
- H_1 : There is a consistent lead-lag relationship at shift k , indicated by a statistically significant average correlation.
 $mean_t(\rho_{n \rightarrow s}^{(t)}(k)) \neq 0$ or $mean_t(\rho_{s \rightarrow n}^{(t)}(k)) \neq 0$
or $mean_t(\rho_{r \rightarrow s}^{(t)}(k)) \neq 0$ or $mean_t(\rho_{s \rightarrow r}^{(t)}(k)) \neq 0$

Key Findings: The Appendix Table 7 and 8 show that **data-driven and fiscal-related sentiment types generally exhibit higher and more significant lead-lag correlations with stock price movements compared to opinion-based sentiment**, particularly in the stock-to-news direction, where coefficients frequently exceed 0.3–0.5 for stocks like TSLA, SPY, and NVDA, while opinion-based correlations are weaker and less consistent across all directions.

5.3.3 Granger Causality Test. Granger Causality Tests determine if one time series can predict another by checking if past values of sentiment indexes for each sentiment type (S_{news} or S_{reddit}) improve the prediction of the current value of another series (S_{stock}), in comparison to predictions that rely solely on the past values of S_{stock} .

- H_0 : The unrestricted model g_k^{news}/g_k^{reddit} cannot improve prediction compared to the restricted model f_k .
(S_{news} or S_{reddit} does not Granger-cause S_{stock})
- H_1 : The unrestricted model g_k^{news}/g_k^{reddit} significantly improves prediction compared to the restricted model f_k .
(S_{news} or S_{reddit} Granger-causes S_{stock})

Key Findings: In the Appendix Table 9, **the Granger Causality Tests show that sentiment indexes (news and Reddit) have limited predictive power for stock price movements across all stocks for all sentiment types**. Due to the inconsistent results of different sentiment types for each stock, we cannot draw any conclusions about the Granger causality.

Conclusion of Hypothesis 3: The analysis demonstrates that data-driven and fiscal-related sentiments in financial headlines have a stronger and more consistent impact on stock price movements compared to opinion-based sentiments, with significant cross-correlations and lead-lag relationships for stocks like TSLA, SPY, and NVDA. However, the Granger Causality Tests indicate limited predictive power across all sentiment types.

6 Experiments

In previous sections, we demonstrated that our sentiment indexes do not show clear predictive power due to the mixed results of the statistical tests. However, we remain interested in exploring the potential usefulness of the indexes for modelling purposes. In this section, we will conduct experiments to determine whether the indexes can improve the performance of our models.

First, we need to define our target for the models. A simple option would be to use the next day’s stock price movement. However, short-term stock price movements are close to a random walk, whereas long-term stock price movements tend to have a higher signal-to-noise ratio. Hence, we chose to adopt the stock price

Table 6: Accuracy Across Models and Feature Combinations

Features	ARs	MLPs (2 layers)	MLPs (3 layers)	Transformers (1 layer)	Transformers (2 layers)	Transformers (3 layers)	Average (excluding ARs)
S_{stock}	0.324 ± 0.004	0.349 ± 0.006	0.344 ± 0.008	0.344 ± 0.010	0.348 ± 0.008	0.348 ± 0.012	0.345 ± 0.009
S_{news}	0.289 ± 0.004	0.345 ± 0.010	0.354 ± 0.014	0.355 ± 0.012	0.366 ± 0.012	0.358 ± 0.010	0.356 ± 0.012
S_{reddit}	0.308 ± 0.004	0.314 ± 0.011	0.332 ± 0.015	0.320 ± 0.014	0.332 ± 0.013	0.329 ± 0.011	0.327 ± 0.013
$S_{stock} + S_{news}$	0.301 ± 0.005	0.349 ± 0.014	0.360 ± 0.014	0.369 ± 0.012	0.362 ± 0.010	0.366 ± 0.009	0.362 ± 0.011
$S_{stock} + S_{reddit}$	0.323 ± 0.005	0.332 ± 0.011	0.340 ± 0.011	0.343 ± 0.009	0.342 ± 0.012	0.341 ± 0.013	0.341 ± 0.011
$S_{news} + S_{reddit}$	0.281 ± 0.003	0.354 ± 0.011	0.362 ± 0.017	0.364 ± 0.014	0.378 ± 0.012	0.373 ± 0.013	0.366 ± 0.013
$S_{stock} + S_{news} + S_{reddit}$	0.319 ± 0.003	0.362 ± 0.011	0.361 ± 0.011	0.360 ± 0.015	0.366 ± 0.015	0.365 ± 0.012	0.361 ± 0.013
Average	0.307 ± 0.004	0.344 ± 0.011	0.350 ± 0.013	0.351 ± 0.012	0.356 ± 0.012	0.354 ± 0.012	

movements over the next five trading days. Still, the continuous values of stock prices can be volatile and noisy, often resulting in poor model performance. In the stock prediction literature [10], the researchers tend to convert the stock price movements as the multiclass classification.

$$q_t = \begin{cases} 0 & \text{if } r_t < \text{Quantile}_{0.333}(r_t) \\ 1 & \text{if } \text{Quantile}_{0.333}(r_t) \leq r_t \leq \text{Quantile}_{0.667}(r_t) \\ 2 & \text{if } r_t > \text{Quantile}_{0.667}(r_t) \end{cases}$$

where $r_t = \frac{P_{t+5} - P_t}{P_t}$ is the rate of return over five days. By using multiclass classification, the models can focus on learning the directions of price movement rather than fitting the precise magnitudes and noises. The multiclass classification also aligns well with our sentiment indexes, which also categorize values into ups, neutrals, and downs. Therefore, this method improves performance and simplifies evaluation.

We evaluate the effectiveness of stock sentiment indexes using three different models: autoregressive models (ARs), multilayer perceptrons (MLPs), and Transformers [8]. For the AR models, we utilize stock price movements and sentiment indexes from the past 15 trading days as inputs. For MLPs, we use multiple hidden layers consisting of 16 units with ReLU activations. For the Transformers, we follow the iTransformers architecture [5], which includes one linear encoding layer, multiple attention layers and one linear prediction layer. Each model takes three time-series features as inputs: stock price movements, news sentiment indexes, and Reddit sentiment indexes. All models will use data from the last 15 trading days. To address the noise in daily values, we apply a past five-day moving average to smooth all time-series curves for the inputs of each model. Our training dataset spans from January 1, 2024, to June 30, 2024, for all six stocks: AAPL, META, NVDA, PLTR, TSLA, and SPY. Our testing dataset ranges from August 1, 2024, to April 14, 2025, avoiding any potential data leakage between the training and testing dataset. All models are trained using the AdamW optimizer [6] with a learning rate of $1e-4$. We report the average and standard deviation from 20 runs for each experiment in Table 6.

6.1 Results

Predicting stock price movements is a highly challenging task. In Table 6, the experiments with Autoregressive (AR) models showed less than 33.33% accuracy, which is even worse than random guessing. One major limitation of the AR model is that it is fundamentally

linear, which can limit its effectiveness in complex tasks like predicting the stock price movements. Additionally, stock price movements often violate the stationarity assumption, which is essential for AR models. In contrast, non-linear models such as MLPs and Transformers significantly outperform AR models. They are better suited to capture the complex patterns of stock price movements. Overall, Transformers performs the best as it is good at learning the interactions of different features through the attention layers.

Moreover, we analyze the effectiveness of our stock sentiment indexes. Our results indicate that relying solely on the Reddit sentiment index S_{reddit} yields the poorest performance, which is consistent with our Granger Causality Tests. Surprisingly, the news sentiment index S_{news} performs better than stock price movements S_{stock} when used as a standalone input feature. We hypothesize that stock prices do not fully reflect the information provided by news, whereas historical prices tend to be backward-looking. However, further investigation is needed to fully understand this phenomenon. Combining any of these three features significantly enhances performance across all MLPs and Transformers experiments. The best results occur when both the news sentiment index and the Reddit sentiment index, $S_{news} + S_{reddit}$, are used together. Therefore, our news and Reddit sentiment indexes prove to be highly valuable for stock prediction models.

6.2 Backtesting

To further evaluate the practical utility of our sentiment indexes, we conducted a backtest simulating a trading strategy based on the model we trained in Table 6. We chose the best model, which is a Transformer with two attention layers, incorporating both news sentiment indexes and stock sentiment indexes. Our trading strategy involves buying and holding stocks when the model predicts a neutral or upward movement in stock prices, and selling the stocks when it predicts a downward movement. For simplicity, we assume there are no transaction fees. We compared the results of our strategy against two baseline strategies: Buy & Hold SPY and Buy & Hold all six stocks equally. In Figure 2, our trading strategy has a 1.123% return over the challenging period, which is better than the two baseline strategies. The result is very impressive because we only trained the model with the first half of 2024 data, which is a completely different trading atmosphere than the current period.

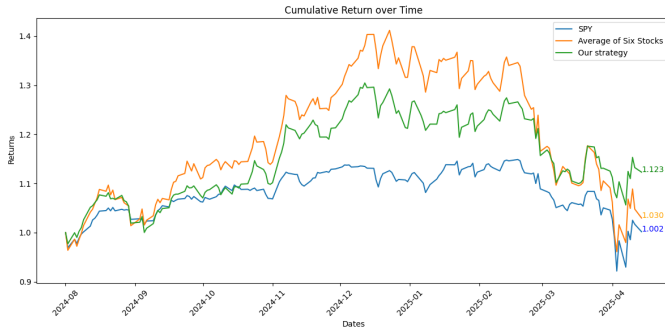


Figure 2: Cumulative Return over Time for SPY, Average of Six Stocks, and Our Strategy.

7 Conclusion

This study indicates that sentiment derived from news headlines and Reddit post titles can be utilized to analyze fluctuations in stock prices. News sentiment, particularly data-driven and fiscal-related categories, shows more robust and reliable correlations with stock prices than sentiment from Reddit. By using these sentiment indexes in machine learning models like Transformers, we improved predictions of stock price directions, achieving a 1.123% return during a challenging period, based on our backtested trading strategy that outperformed simpler trading strategies. However, the study has limitations: it only covers a short period (March 2024–April 2025), which may miss longer trends; historical data are hard and expensive to access, especially Reddit data, reducing the dataset’s size; the reliance on the DeepSeek model [2] for sentiment classification may introduce biases; and choosing specific sources like Nasdaq, Benzinga, and Reddit may skew results, as different platforms express different sentiments. Future work should aim to include more historical data if accessibility and cost issues can be resolved. Testing other sentiment analysis tools, like BERT [3], could improve accuracy and avoid model biases. Additionally, it is important to study how market reactions can provide insights into the role of sentiment in market behavior. For instance, phenomena such as the fear of missing out (FOMO) or overreacting to financial events can lead to significant price fluctuations. By understanding these dynamics, we can enhance the reliability and usefulness of sentiment-based models for financial analysis.

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A Additional Graphs

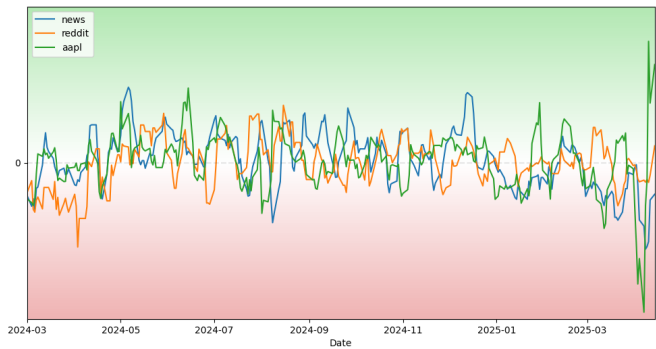


Figure 3: AAPL Sentiment and Price Trends

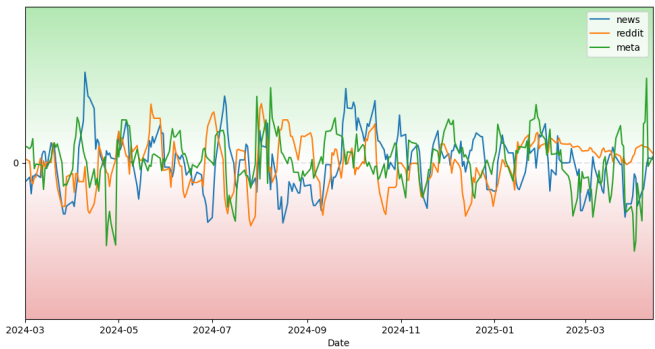


Figure 4: META Sentiment and Price Trends

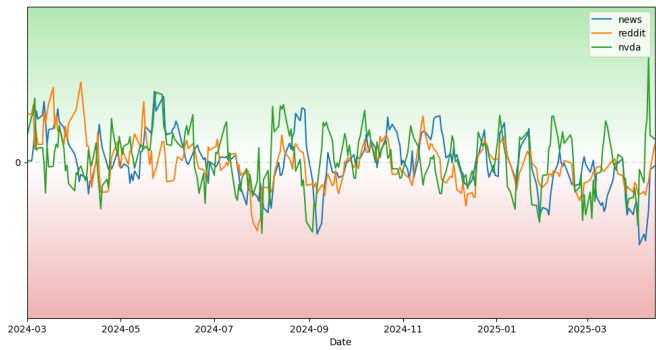


Figure 5: NVDA Sentiment and Price Trends

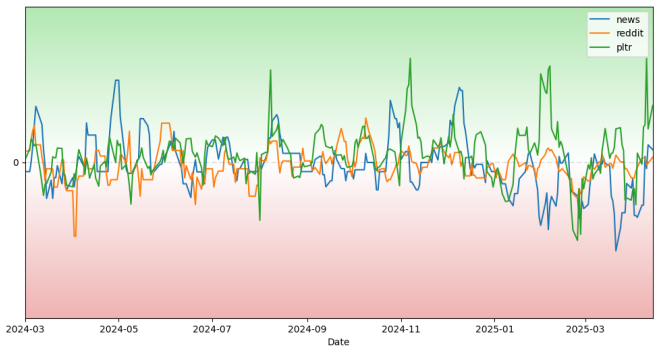


Figure 6: PLTR Sentiment and Price Trends

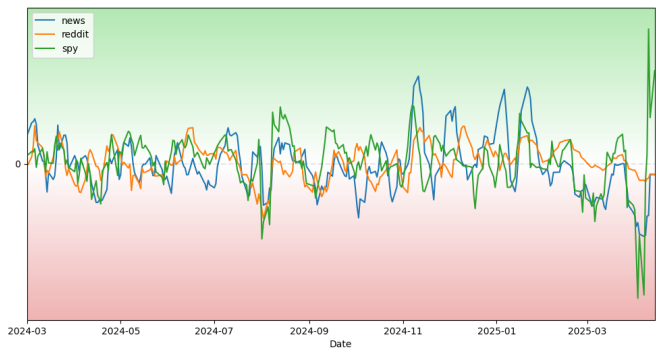


Figure 7: SPY Sentiment and Price Trends

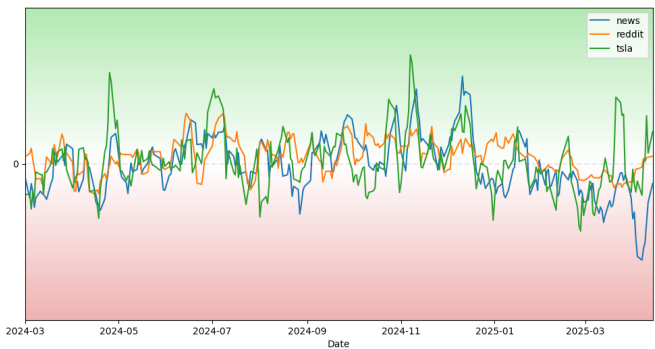


Figure 8: TSLA Sentiment and Price Trends

Figure 9: Sentiment and Price Trends

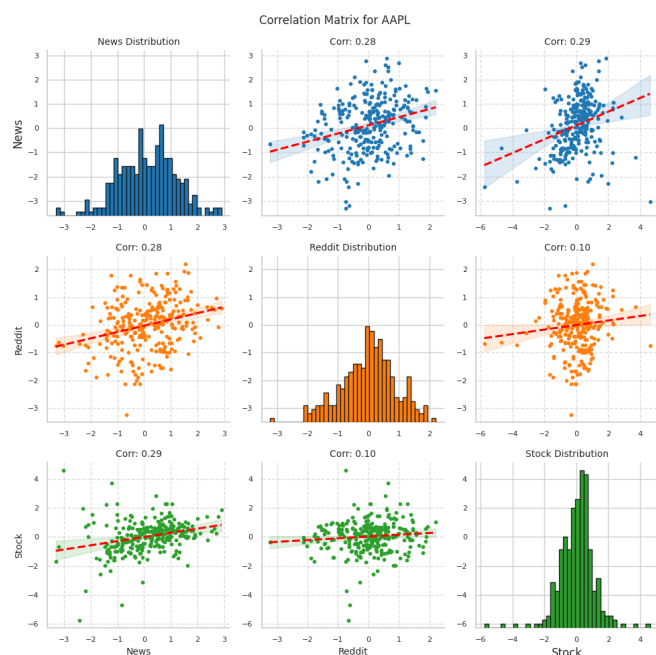


Figure 10: AAPL Correlation Matrix Plots

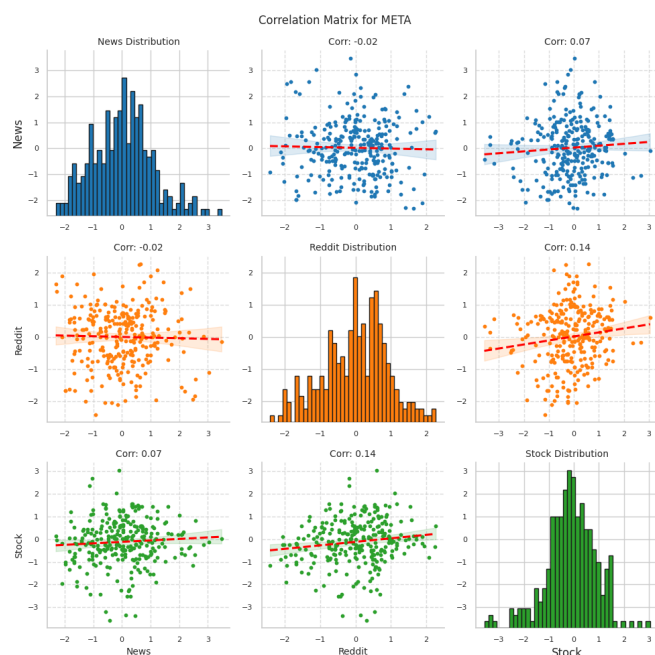


Figure 11: META Correlation Matrix Plots

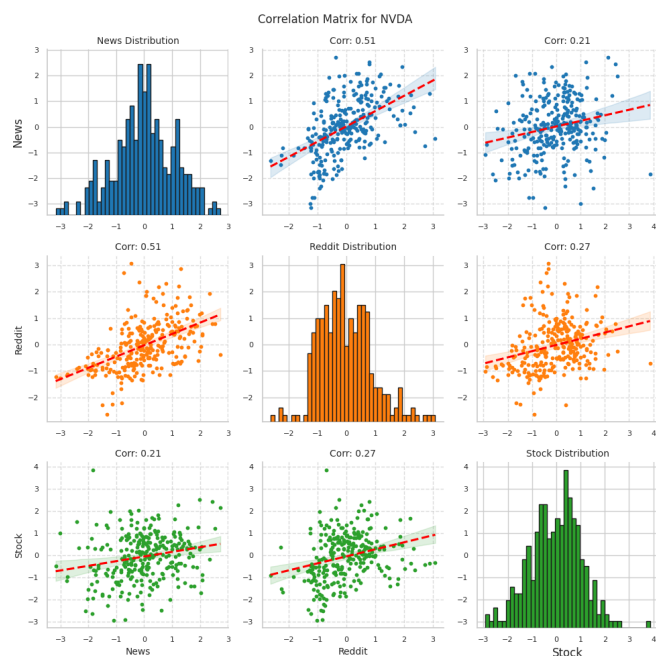


Figure 12: NVDA Correlation Matrix Plots

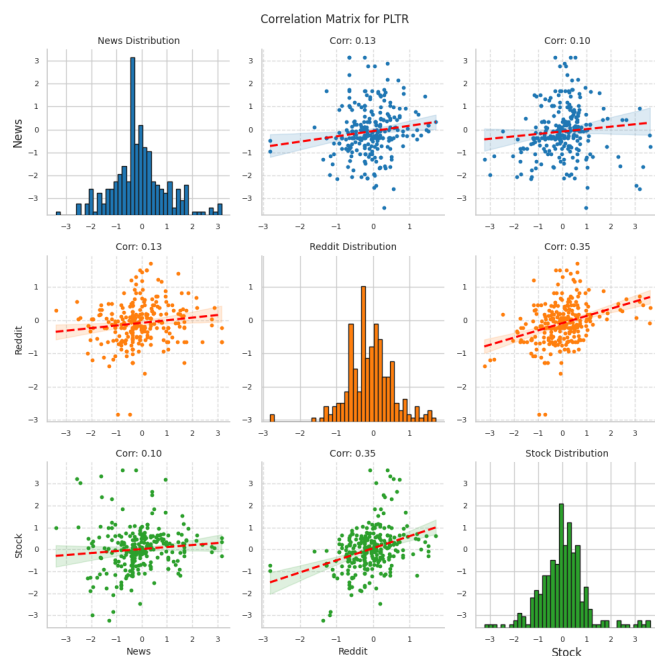


Figure 13: PLTR Correlation Matrix Plots

Figure 14: Correlation Matrix Plots

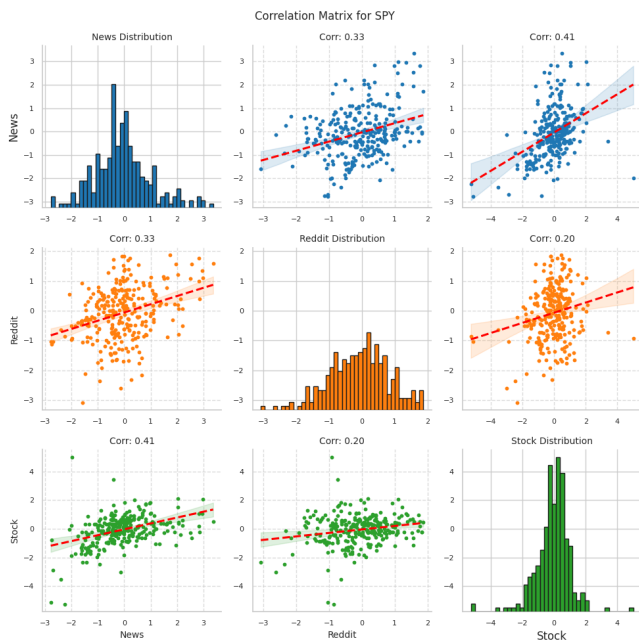


Figure 15: SPY Correlation Matrix Plots

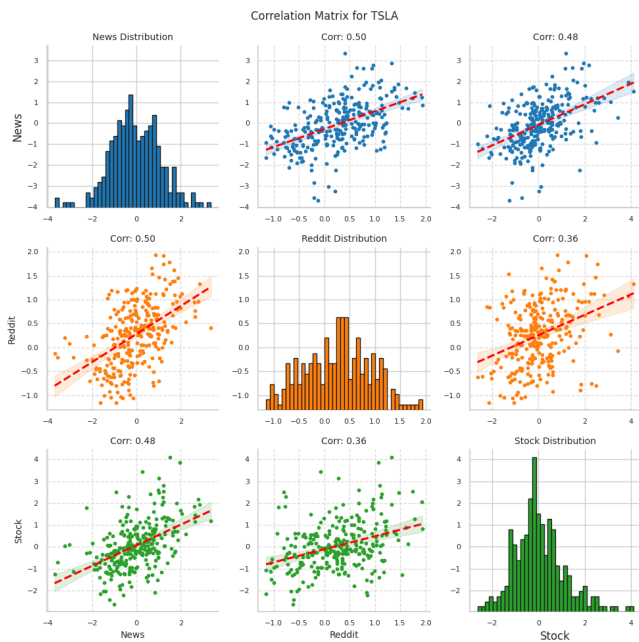


Figure 16: TSLA Correlation Matrix Plots

Figure 17: Correlation Matrix Plots

B Additional Tables

Table 7: Lead-Lag Correlation between Sentiment Indexes and Stock Price Movements for each Sentiment Type for AAPL, META and NVDA

Stocks	Sentiment Type	Direction	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5
AAPL	Data-driven	news→stock	0.19**	0.13**	-0.01	-0.10**	-0.11**
		stock→news	0.35**	0.28**	0.19**	0.13**	0.01
		reddit→stock	-0.13**	-0.18**	-0.21**	-0.23**	-0.25**
		stock→reddit	-0.02	0.05*	0.12**	0.16**	0.20**
	Fiscal-related	news→stock	-0.05	-0.06*	-0.06	-0.04	0.01
		stock→news	0.10**	0.11**	0.12**	0.10**	0.06
		reddit→stock	-0.01	-0.05	-0.09**	-0.15**	-0.18**
		stock→reddit	0.11**	0.12**	0.07**	0.03	0.01
	Opinion-based	news→stock	-0.03	0.01	-0.07*	-0.15**	-0.14**
		stock→news	-0.11**	-0.16**	0.05*	-0.10**	-0.23**
		reddit→stock	-0.01	-0.05	-0.04	-0.08*	-0.12**
		stock→reddit	-0.06*	-0.03	0.06**	0.12**	0.14**
META	Data-driven	news→stock	-0.09**	-0.11**	-0.16**	-0.06*	0.05
		stock→news	0.11**	0.19**	0.17**	0.15**	0.10**
		reddit→stock	0.19**	0.16**	0.13**	0.10**	0.04
		stock→reddit	0.06*	-0.03	-0.09**	-0.13**	-0.07**
	Fiscal-related	news→stock	0.03	-0.03	-0.03	0.05	0.09**
		stock→news	0.13**	0.16**	0.08**	0.04	-0.01
		reddit→stock	0.09**	0.03	-0.07*	-0.11**	-0.06*
		stock→reddit	0.19**	0.17**	0.10**	0.06*	0.01
	Opinion-based	news→stock	-0.02	-0.02	-0.02	0.01	0.02
		stock→news	-0.08**	0.01	-0.01	0.03**	-0.05*
		reddit→stock	-0.11**	-0.13**	-0.09**	-0.04	-0.06*
		stock→reddit	0.01	-0.02	0.02	-0.00	-0.01
NVDA	Data-driven	news→stock	0.00	-0.04	-0.10**	-0.12**	-0.06
		stock→news	0.32**	0.32**	0.22**	0.15**	0.05*
		reddit→stock	0.03	-0.00	-0.05*	-0.05*	-0.07**
		stock→reddit	0.17**	0.23**	0.19**	0.21**	0.20**
	Fiscal-related	news→stock	-0.08**	-0.21**	-0.30**	-0.25**	-0.13**
		stock→news	0.20**	0.28**	0.23**	0.13**	0.13**
		reddit→stock	-0.09**	-0.18**	-0.22**	-0.23**	-0.22**
		stock→reddit	0.15**	0.16**	0.14**	0.08**	0.01
	Opinion-based	news→stock	-0.11**	-0.04	0.02	0.16**	0.19**
		stock→news	-0.01	0.07	0.22**	0.27**	0.19**
		reddit→stock	0.20**	0.14**	0.07	0.01	-0.06
		stock→reddit	0.27**	0.31**	0.32**	0.27**	0.19**

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).

Table 8: Lead-Lag Correlation between Sentiment Indexes and Stock Price Movements for each Sentiment Type for PLTR, TSLA and SPY

Stocks	Sentiment Type	Direction	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5
PLTR	Data-driven	news→stock	-0.06**	-0.20**	-0.23**	-0.15**	-0.01
		stock→news	0.21**	0.25**	0.19**	0.04	-0.02
		reddit→stock	-0.09**	-0.15**	-0.13**	-0.04	0.01
		stock→reddit	0.02	0.02	-0.03	-0.03	0.03
	Fiscal-related	news→stock	0.17**	0.08**	-0.03	-0.09**	-0.05
		stock→news	0.23**	0.16**	0.01	-0.11**	-0.17**
		reddit→stock	0.11**	0.06*	0.01	-0.02	-0.03
		stock→reddit	0.20**	0.20**	0.16**	0.13**	0.01
	Opinion-based	news→stock	-0.19**	-0.11**	-0.13**	0.01	-0.02
		stock→news	-0.09**	0.00	0.08**	0.08**	0.03*
		reddit→stock	0.08**	0.10**	0.10**	0.08*	0.02
		stock→reddit	0.07**	0.06**	0.05**	0.03*	0.08**
TSLA	Data-driven	news→stock	0.12**	0.07*	-0.01	-0.09**	-0.06*
		stock→news	0.41**	0.43**	0.40**	0.33**	0.18**
		reddit→stock	0.10**	-0.00	-0.05	-0.10**	-0.10**
		stock→reddit	0.28**	0.30**	0.29**	0.19**	0.03
	Fiscal-related	news→stock	0.15**	0.04	-0.12**	-0.14**	-0.10**
		stock→news	0.49**	0.49**	0.33**	0.20**	0.01
		reddit→stock	0.02	-0.08**	-0.19**	-0.24**	-0.25**
		stock→reddit	0.29**	0.32**	0.23**	0.23**	0.14**
	Opinion-based	news→stock	0.07	0.05*	-0.11**	-0.07**	0.05**
		stock→news	0.01	0.15**	-0.02	-0.10**	-0.03
		reddit→stock	0.14**	0.10**	0.04	-0.02	-0.06*
		stock→reddit	0.24**	0.18**	0.12**	0.12**	0.12**
SPY	Data-driven	news→stock	0.08**	-0.07**	-0.21**	-0.31**	-0.27**
		stock→news	0.42**	0.37**	0.26**	0.19**	0.13**
		reddit→stock	0.03	-0.04	-0.09**	-0.17**	-0.19**
		stock→reddit	0.14**	0.13**	0.11**	0.09**	0.06*
	Fiscal-related	news→stock	0.15**	0.04	-0.13**	-0.22**	-0.18**
		stock→news	0.52**	0.51**	0.39**	0.30**	0.20**
		reddit→stock	-0.00	-0.05	-0.05	-0.06	-0.01
		stock→reddit	0.07**	0.05	0.04	0.01	0.02
	Opinion-based	news→stock	-0.07**	0.13**	0.13**	0.03	0.01
		stock→news	-0.11**	-0.04	-0.13**	-0.03	0.03
		reddit→stock	-0.01	0.01	0.06	0.10*	0.12*
		stock→reddit	0.02*	0.05**	0.07**	0.01	-0.03

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).

Table 9: P-value of the Granger Causality Test for each Sentiment Type

Stocks	Sentiment Type	Test	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
AAPL	Data-driven	f_k vs g_k^{news}	0.67	0.96	0.04*	0.05*	0.05*
		f_k vs g_k^{reddit}	0.78	0.97	0.98	0.97	0.97
	Fiscal-related	f_k vs g_k^{news}	0.35	0.33	0.47	0.46	0.31
		f_k vs g_k^{reddit}	0.36	0.82	0.92	0.72	0.74
	Opinion-based	f_k vs g_k^{news}	0.88	0.99	0.99	0.96	0.58
		f_k vs g_k^{reddit}	0.63	0.86	0.93	0.43	0.45
META	Data-driven	f_k vs g_k^{news}	0.17	0.33	0.14	0.09	0.12
		f_k vs g_k^{reddit}	0.04*	0.07	0.12	0.10	0.11
	Fiscal-related	f_k vs g_k^{news}	0.67	0.61	0.69	0.54	0.35
		f_k vs g_k^{reddit}	0.57	0.24	0.23	0.53	0.41
	Opinion-based	f_k vs g_k^{news}	0.61	0.89	0.97	0.30	0.44
		f_k vs g_k^{reddit}	0.48	0.27	0.51	0.65	0.63
NVDA	Data-driven	f_k vs g_k^{news}	0.17	0.49	0.28	0.78	0.60
		f_k vs g_k^{reddit}	0.53	0.85	0.65	0.82	0.89
	Fiscal-related	f_k vs g_k^{news}	0.04*	0.00**	0.01**	0.11	0.13
		f_k vs g_k^{reddit}	0.01*	0.07	0.13	0.29	0.33
	Opinion-based	f_k vs g_k^{news}	0.43	0.76	0.78	0.78	0.63
		f_k vs g_k^{reddit}	0.61	0.70	0.90	0.78	0.25
PLTR	Data-driven	f_k vs g_k^{news}	0.27	0.32	0.23	0.02*	0.01*
		f_k vs g_k^{reddit}	0.76	0.85	0.61	0.71	0.67
	Fiscal-related	f_k vs g_k^{news}	0.48	0.33	0.48	0.68	0.59
		f_k vs g_k^{reddit}	0.46	0.44	0.64	0.72	0.84
	Opinion-based	f_k vs g_k^{news}	0.96	0.67	0.68	0.61	0.80
		f_k vs g_k^{reddit}	0.05*	0.12	0.19	0.16	0.07
TSLA	Data-driven	f_k vs g_k^{news}	0.76	0.56	0.49	0.26	0.04*
		f_k vs g_k^{reddit}	0.97	0.74	0.35	0.34	0.19
	Fiscal-related	f_k vs g_k^{news}	0.49	0.86	0.35	0.10	0.01**
		f_k vs g_k^{reddit}	0.56	0.97	0.98	0.97	0.88
	Opinion-based	f_k vs g_k^{news}	0.02*	0.07	0.27	0.46	0.47
		f_k vs g_k^{reddit}	0.72	0.05*	0.05	0.14	0.21
SPY	Data-driven	f_k vs g_k^{news}	0.03*	0.06	0.02*	0.10	0.02*
		f_k vs g_k^{reddit}	0.78	0.76	0.38	0.33	0.35
	Fiscal-related	f_k vs g_k^{news}	0.05	0.50	0.08	0.65	0.17
		f_k vs g_k^{reddit}	0.62	0.80	0.87	0.86	0.86
	Opinion-based	f_k vs g_k^{news}	0.83	0.90	0.69	0.89	0.44
		f_k vs g_k^{reddit}	0.97	0.66	0.60	0.55	0.12

Note: * indicates p-value < 0.05, ** indicates p-value < 0.01. Coefficients without a star are not significant (p-value ≥ 0.05).